

Glossary

1-repetition maximum (1RM)—The maximal amount of weight that can be lifted just one time.

1RM—See 1-repetition maximum.

acclimation (heat acclimation)—Physiological adaptation to repeated environmental stresses, occurring over a relatively brief period of time (days to weeks). Acclimation often occurs in a laboratory environment.

acclimatization—Physiological adaptation to repeated environmental stress in a natural environment, occurring over months and years of living and exercising in that environment.

ACE—See angiotensin-converting enzyme.

acetylcholine—A primary neurotransmitter that transmits impulses across the synaptic cleft.

acetyl CoA—See acetyl coenzyme A.

acetyl coenzyme A (acetyl CoA)—The compound that forms the common entry point into the Krebs cycle for the oxidation of carbohydrate and fat.

actin—A thin protein filament that acts with myosin filaments to produce muscle action.

action potential—A rapid and substantial depolarization of the membrane of a neuron or muscle cell that is conducted through the cell.

activation energy—The initial energy required to start a chemical reaction or chain of reactions.

acute altitude (mountain) sickness—Illness characterized by headache, nausea, vomiting, dyspnea, and insomnia. It typically begins 6 to 96 h after one reaches high altitude and lasts several days.

acute exercise—A single bout of exercise.

acute muscle soreness—Soreness or pain felt during and immediately after an exercise bout.

acute overload—An average training load, whereby the athlete is stressing the body to the extent necessary to improve both physiological function and performance.

adenosine diphosphate (ADP)—A high-energy phosphate compound from which ATP is formed.

adenosine triphosphatase (ATPase)—An enzyme that splits the last phosphate group off ATP, releasing a large amount of energy and reducing the ATP to ADP

and P_i .

adenosine triphosphate (ATP)—A high-energy phosphate compound from which the body derives its energy.

ADH—See antidiuretic hormone.

adipokines—Hormones that are released by, or act on, fat tissue.

adolescence—The period of life between the end of childhood and the beginning of adulthood. The onset of puberty marks the beginning of adolescence.

ADP—See adenosine diphosphate.

adrenaline—A chemical compound that serves as a neurotransmitter throughout the body. Also called epinephrine.

adrenergic—Refers to norepinephrine or epinephrine (also called noradrenaline and adrenaline, respectively).

aerobic metabolism—A process occurring in the mitochondria that uses oxygen to produce energy (ATP). Also known as cellular respiration.

aerobic power—Another term for maximal oxygen uptake, or $\dot{V}O_{2\max}$.

aerobic training—Training that improves the efficiency of the aerobic energy-producing systems and can improve cardiorespiratory endurance.

afterload—The pressure against which the heart must pump blood, determined by the peripheral resistance in the large arteries.

air plethysmography—A procedure for assessing body composition by using air displacement to measure body volume, allowing the calculation of body density.

aldosterone—A mineralocorticoid hormone secreted by the adrenal cortex that prevents dehydration by promoting renal absorption of sodium.

α -motor neuron—A neuron innervating extrafusal skeletal muscle fibers.

alveoli—Grapelike clusters, or air sacs, at the ends of the terminal bronchioles.

amenorrhea—The absence (primary amenorrhea) or cessation (secondary amenorrhea) of normal menstrual function.

amphetamine—A central nervous system stimulant proposed to have ergogenic properties.

anabolic steroids—Prescription drugs with the anabolic (growth stimulating) characteristics of testosterone, taken by some athletes to increase body size, muscle mass, and strength.

anaerobic metabolism—The production of energy (ATP) in the absence of oxygen.

anaerobic power—Mean or peak power output in exercise lasting 30 s or less.

anaerobic threshold—The point at which the metabolic demands of exercise can no longer be met by available aerobic sources and at which an increase in anaerobic metabolism occurs, reflected by an increase in blood lactate concentration.

anaerobic training—Training that improves the efficiency of the anaerobic energy-producing systems and can increase muscular strength and tolerance for

acid–base imbalances during high-intensity effort.

angiotensin-converting enzyme (ACE)—An enzyme that converts angiotensin I to angiotensin II.

anorexia nervosa—A clinical eating disorder characterized by distorted body image, intense fear of fatness or weight gain, amenorrhea, and refusal to maintain more than the minimal normal weight based on age and height.

antidiuretic hormone (ADH)—A hormone secreted by the pituitary gland that regulates fluid and electrolyte balance in the blood by reducing urine production.

arginine vasopressin—See antidiuretic hormone.

arterial–mixed venous oxygen difference, or $(a-\bar{v})O_2$ difference—The difference in oxygen content between arterial and mixed venous blood, which reflects the amount of oxygen removed by the whole body.

arterial–venous oxygen difference, or $(a-v)O_2$ difference—The difference in oxygen content between arterial and venous blood at the tissue level.

arteries—Blood vessels that transport blood away from the heart.

arterioles—The smallest arteries that transport blood from larger arteries to the capillaries.

atherosclerosis—A condition that involves changes in the lining of the arteries and plaque accumulation, leading to progressive narrowing of the arteries.

athlete's heart—A nonpathological enlarged heart, often found in endurance athletes, that results primarily from left ventricular hypertrophy in response to training.

ATP—See adenosine triphosphate.

ATPase—See adenosine triphosphatase.

ATP-PCr system—The short-term anaerobic energy system that maintains ATP levels. Breakdown of phosphocreatine (PCr) frees P_i , which then combines with ADP to form ATP.

atrioventricular (AV) node—The specialized mass of conducting cells in the heart located at the atrioventricular junction.

atrophy—Loss of size, or mass, of body tissue, such as muscle atrophy with disuse.

autocrine—A locally acting messenger molecule, such as a prostaglandin.

autogenic inhibition—Reflex inhibition of a motor neuron in response to excessive tension in the muscle fibers it supplies, as monitored by the Golgi tendon organs.

AV node—See atrioventricular node.

$(a-v)O_2$ difference—See arterial–venous oxygen difference.

$(a-\bar{v})O_2$ difference—See arterial–mixed venous oxygen difference.

axon hillock—A part of the neuron, between the cell body and the axon, that controls traffic down the axon through summation of excitatory and inhibitory postsynaptic potentials.

axon terminal—One of numerous branched endings of an axon. Also known as a terminal fibril.

barometric pressure (P_b)—The total pressure exerted by the atmosphere at a given altitude.

baroreceptor—Stretch receptor located within the cardiovascular system that senses changes in blood pressure.

basal metabolic rate (BMR)—The lowest rate of body metabolism (energy use) that can sustain life, measured after an overnight sleep in a laboratory under optimal conditions of quiet, rest, and relaxation and after a 12 h fast. *See also* resting metabolic rate.

β -blockers—A class of drugs that block transmission of neural impulses from the sympathetic nervous system, proposed to have ergogenic properties.

β -cells—Cells in the islets of Langerhans in the pancreas that secrete insulin.

β -oxidation—The first step in fatty acid oxidation, in which fatty acids are broken into separate two carbon units of acetic acid, each of which is then converted to acetyl CoA.

bioelectric impedance—A procedure for assessing body composition in which an electrical current is passed through the body. The resistance to current flow through the tissues reflects the relative amount of fat present.

bioenergetics—Term given to the study of metabolic processes that yield or consume energy.

bioinformatics—The science of analyzing complex biological data such as genetic codes.

blood doping—Any means by which a person's total volume of red blood cells is increased, typically via transfusion of red blood cells or use of erythropoietin.

blood lipids—Blood-borne fats, such as triglycerides and cholesterol.

BMI—See body mass index.

BMR—See basal metabolic rate.

body composition—The chemical composition of the body. The model used in this book considers two components: fat-free mass and fat mass.

body density—Body weight divided by body volume.

body mass index (BMI)—A measurement of body overweight or obesity determined by dividing weight (in kilograms) by height (in meters) squared. BMI is highly correlated with body composition.

bone mineral density—The mass of bone per unit volume. Decreased bone mineral density increases the risk of fractures.

Borg RPE scale—A numerical scale for rating perceived exertion.

Boyle's gas law—Law stating that at a constant temperature, the number of gas molecules in a given volume depends on the pressure.

bradycardia—A resting heart rate lower than 60 beats/min.

bulimia nervosa—A clinical eating disorder characterized by recurrent episodes of binge eating, a feeling of lack of control during these binges, and purging

behavior, which may include self-induced vomiting and use of laxatives and diuretics. Sometimes the disorder also includes fasting or excessive exercise behaviors.

CAD—See coronary artery disease.

caffeine—A central nervous system stimulant believed by some athletes to have ergogenic properties.

calorie (cal)—A unit of measure of energy in biological systems, where 1 calorie is equal to the amount of heat energy needed to raise the temperature of 1 g of water 1 °C, from 15 to 16 °C.

calorimeter—A device for measuring the heat produced by the body (or by specific chemical reactions).

cAMP—See cyclic adenosine monophosphate.

capillaries—The smallest vessels transporting blood from the heart to the tissues and the actual sites of exchange between the blood and tissue.

capillary-to-fiber ratio—The number of capillaries per muscle fiber.

carbohydrate—An organic compound formed from carbon, hydrogen, and oxygen; includes starches, sugars, and cellulose.

carbohydrate loading—Increased dietary consumption of carbohydrate, a process used by athletes to increase carbohydrate stores in the body before prolonged endurance exercise.

cardiac cycle—The period that includes all events between two consecutive heartbeats.

cardiac hypertrophy—Enlargement of the heart by increases in muscle wall thickness or chamber size or both.

cardiac output ($\dot{Q}$)—The volume of blood pumped out by the heart per minute. $\dot{Q}$ = heart rate $\times$ stroke volume.

cardiorespiratory endurance—The ability of the body to sustain prolonged exercise.

cardiovascular deconditioning—A decrease in the cardiovascular system's ability to deliver sufficient oxygen and nutrients.

cardiovascular drift—An increase in heart rate during exercise to compensate for a decrease in stroke volume. This compensation helps maintain a constant cardiac output.

catabolism—The tearing down of body tissue; the destructive phase of metabolism.

catecholamines—Biologically active amines (organic compounds derived from ammonia), such as epinephrine and norepinephrine, that serve as neurotransmitters in the sympathetic nervous system and as hormones.

central command—Information originating in the brain that is transmitted to the cardiovascular, muscular, or pulmonary systems.

central governor theory—Theory proposing that processes occur in the brain that regulate power output by the muscles to prevent unsafe levels of exertion.

central nervous system (CNS)—System consisting of the brain and spinal cord.

chemoreceptor—A sensory organ capable of reacting to a chemical stimulus.

Cheyne-Stokes breathing—Alternating periods of rapid breathing and slow, shallow breathing, including periods in which breathing may actually cease temporarily. A symptom of acute mountain sickness.

childhood—The period of life between the first birthday and the onset of puberty.

cholecystikinin (CCK)—A hormone released by the gastrointestinal tract that signals the brain to suppress hunger.

cholinergic—Systems mediated by the neurotransmitter acetylcholine.

chronic adaptation—A physiological change that occurs when the body is exposed to repeated exercise bouts over weeks or months. These changes generally improve the body's efficiency at rest and during exercise.

chronic hypertrophy—An increase in muscle size that results from repeated long-term resistance training.

CNS—See central nervous system.

cold habituation—A response to repeated cold exposure, often to the hands and face, in which skin vasoconstrictor and shivering responses are blunted.

concentric contraction—Muscle shortening.

conduction (K)—(1) Transfer of heat through direct molecular contact with a solid object; (2) movement of an electrical impulse, such as through a neuron.

congenital heart disease—A heart defect present at birth that occurs from abnormal prenatal development of the heart or associated blood vessels. Also known as congenital heart defect.

continuous training—Training at a moderate to high intensity without stopping to rest.

control group—In an experimental design, the nontreated group to which the experimental group is compared.

convection (C)—The transfer of heat or cold via the movement of a gas or liquid across an object, such as the body.

coronary artery disease (CAD)—A disease characterized by pathological changes in the coronary arteries that supply blood flow to the myocardium.

cortisol—A corticosteroid hormone released from the adrenal cortex that stimulates gluconeogenesis, increases mobilization of free fatty acids, decreases use of glucose, and stimulates catabolism of protein. Also known as hydrocortisone.

creatine—A substance found in skeletal muscles most commonly in the form of PCr. Creatine supplements are often used as ergogenic aids because they are theorized to increase PCr levels, thus enhancing the ATP-PCr energy system by better maintaining muscle ATP levels.

creatine kinase—The enzyme that facilitates the breakdown of PCr to creatine and P_i .

critical power—The maximum power output or exercise intensity that can be sustained, at least theoretically, without exhaustion occurring.

critical temperature theory—Theory that prolonged exercise in hot environments is limited by attainment of a fixed elevated core temperature.

crossover concept—The exercise intensity at which fat and carbohydrate utilization intersect as the energy derived from fat decreases and the energy from carbohydrate increases.

crossover design—Experimental design in which the control group becomes the experimental group after the first experimental period, and vice versa.

cross-sectional research design—A research design in which a cross section of a population is tested at one specific time and then data from groups within that population are compared.

cross-training—Training for more than one sport at the same time, or training multiple fitness components (such as endurance, strength, and flexibility) within the same period.

cycle ergometer—An exercise device that uses cycling to measure physical work.

cyclic adenosine monophosphate (cAMP)—Intracellular second messenger that mediates hormone action.

cytochrome—A series of iron-containing proteins that facilitate the transport of electrons within the electron transport chain.

Dalton's law—A principle stating that the total pressure exerted by a mixture of gases is equal to the sum of the partial pressures of those individual gases.

DBP—See diastolic blood pressure.

dead space—The volume of air inhaled that does not take part in the gas exchange because it remains in the conducting airways.

dehydration—Loss of body fluids.

delayed-onset muscle soreness (DOMS)—Muscle soreness that develops a day or two after a heavy bout of exercise and is associated with actual injury within the muscle.

densitometry—The measurement of body density.

dependent variable—The physiological factor that is allowed to vary as another factor (the independent variable) is manipulated. Usually plotted on the y-axis.

depolarization—A decrease in the electrical potential across a membrane, as when the inside of a neuron becomes less negative relative to the outside.

detraining—Changes in physiological function in response to a reduction or cessation of regular physical training.

development—Changes that occur in the body starting at conception and continuing through adulthood; differentiation along specialized lines of function, reflecting changes that accompany growth.

diabetes mellitus—A disorder of carbohydrate metabolism characterized by hyperglycemia (high blood sugar levels) and glycosuria (presence of sugar in the

urine). The disease develops when there is inadequate production of insulin by the pancreas or inadequate utilization of insulin by the cells.

diastolic blood pressure (DBP)—The lowest arterial pressure, resulting from ventricular diastole (the resting phase).

direct calorimetry—A method that gauges the body's rate and quantity of energy production by direct measurement of the body's heat production.

direct gene activation—The method of action of steroid hormones. They bind to receptors in the cell, and then the hormone–receptor complex enters the nucleus and activates certain genes.

disordered eating—Abnormal eating behavior that ranges from excessive restriction of food intake to pathological behaviors, such as self-induced vomiting and laxative abuse. Disordered eating can lead to clinical eating disorders, such as anorexia nervosa and bulimia nervosa.

diuretics—Substances that promote water excretion.

diurnal variation—Fluctuations in physiological responses that occur during a 24 h period.

DOMS—See delayed-onset muscle soreness.

dose–response relation—A relationship between two variables in which one changes predictably as the other increases or decreases.

downregulation—Decreased cellular sensitivity to a hormone, likely the result of a decreased number of cell receptors available to bind with the hormone.

dry heat exchange—Heat transfer by the combined avenues of convection, conduction, and radiation.

dual-energy X-ray absorptiometry (DEXA)—A technique used to assess both regional and total body composition through the use of X-ray absorptiometry.

dynamic contraction—Any muscle action that produces joint movement.

dyspnea—Labored or difficult breathing.

EAMCs—See exercise-associated muscle cramps.

eating disorders—A group of clinical disorders involving eating. *See also* anorexia nervosa, bulimia nervosa.

eccentric contraction—Any muscle action in which muscle lengthens.

eccentric training—Training that involves eccentric action.

eccrine sweat glands—Simple sweat glands dispersed over the body surface that respond to increases in core or skin temperature (or both) and facilitate thermoregulation.

ECG—See electrocardiogram and exercise electrocardiogram.

EDV—See end-diastolic volume.

EF—See ejection fraction.

effector (efferent) nerves—The motor division of the peripheral nervous system, carrying impulses from the CNS toward the periphery.

EIAH—See exercise-induced arterial hypoxemia.

ejection fraction (EF)—The fraction of blood pumped out of the left ventricle with each contraction, determined by dividing stroke volume by end-diastolic volume and expressed as a percentage.

electrical stimulation—Stimulation of a muscle by passing an electrical current through it.

electrocardiogram (ECG)—A recording of the heart's electrical activity.

electrocardiograph—A machine used to obtain an electrocardiogram.

electrolyte—A dissolved substance that can conduct an electrical current.

electron transport chain—A series of chemical reactions that convert the hydrogen ion generated by glycolysis and the Krebs cycle into water and produce energy for oxidative phosphorylation.

end branches—Branches coming off the ends of the axons leading to the axon terminals.

end-diastolic volume (EDV)—The volume of blood inside the left ventricle at the end of diastole, just before contraction.

endomysium—A sheath of connective tissue that covers each muscle fiber.

endothelial dysfunction—Negative changes in the cells that line the lumen of blood vessels resulting in a relative inability of those vessels to constrict or dilate.

endothelium—Layer of thin cells that line the lumen of the blood vessels.

end-systolic volume (ESV)—The volume of blood remaining in the left ventricle at the end of systole, just after contraction.

environmental physiology—Study of the effects of the environment (heat, cold, altitude, hyperbaria, and so on) on the function of the body.

enzyme—Protein molecules that speed up reactions by lowering their energy of activation.

ephedrine—A sympathomimetic amine derived from ephedra herbs (also known as ma huang) and used as a decongestant and as a bronchodilator in the treatment of asthma.

epigenetics—The study of changes in organisms caused by modifying gene expression rather than changing the genetic code itself.

epimysium—The outer connective tissue that surrounds an entire muscle, holding it together.

epinephrine—A catecholamine released from the adrenal medulla that, along with norepinephrine, prepares the body for a fight-or-flight response. It is also a neurotransmitter. *See also* catecholamines.

EPOC—See excess postexercise oxygen consumption.

EPSP—See excitatory postsynaptic potential.

ergogenic—Able to improve work or performance.

ergolytic—Able to impair work or performance.

ergometer—An exercise device that allows the amount and rate of a person's physical work to be controlled (standardized) and measured.

erythropoietin (EPO)—The hormone that stimulates erythrocyte (red blood cell) production.

essential amino acids—The eight or nine amino acids necessary for human growth that the body cannot synthesize and are thus essential parts of our diets.

estrogen—A female sex hormone.

ESV—See end-systolic volume.

eumenorrhea—Normal menstrual function.

evaporation (E)—Heat loss through the conversion of water (such as in sweat) to vapor.

excessive training—Training in which volume, intensity, or both are too great or are increased too quickly without proper progression.

excess postexercise oxygen consumption (EPOC)—Elevated oxygen consumption above resting levels after exercise; at one time referred to as oxygen debt.

excitation-contraction coupling—The sequence of events by which a nerve impulse reaches the muscle membrane and leads to cross-bridge activity and thus muscle contraction.

excitatory postsynaptic potential (EPSP)—A depolarization of the postsynaptic membrane caused by an excitatory impulse.

exercise-associated muscle cramps (EAMCs)—Painful prolonged contractions of muscles that accompany or result from muscle contractions.

exercise electrocardiogram (ECG)—A recording of the heart's electrical activity during exercise.

exercise-induced arterial hypoxemia (EIAH)—A decline in arterial PO_2 and arterial oxygen saturation during maximal or near-maximal exercise.

exercise physiology—The study of how body structure and function are altered by exposure to acute and chronic bouts of exercise.

exercise prescription—Individualization of the prescription of exercise duration, frequency, intensity, and mode.

expiration—The process by which air is forced out of the lungs through relaxation of the inspiratory muscles and elastic recoil of the lung tissue, which increase the pressure in the thorax.

external respiration—The process of bringing air into the lungs and the resulting exchange of gas between the alveoli and the capillary blood.

extracellular fluid—The 35% to 40% of the water in the body that is outside the cells, including interstitial fluid, blood plasma, lymph, cerebrospinal fluid, and other fluids.

extrinsic neural control—Redistribution of blood at the system or body level through neural mechanisms.

Fartlek training—Developed in the 1930s; the term comes from the Swedish for “speed play.” This type of training combines continuous and interval training and stresses both the aerobic and anaerobic energy pathways.

fascicle—A small bundle of muscle fibers wrapped in a connective tissue sheath within a muscle.

fat—A class of organic compounds with limited water solubility that exists in the body in many forms, such as triglycerides, free fatty acids, phospholipids, and steroids.

fat-free mass—The mass (weight) of the body that is not fat, including muscle, bone, skin, and organs.

fatigue—General sensations of tiredness and accompanying decrements in muscular performance.

fat mass—The absolute amount or mass of body fat.

fatty streaks—Early lipid deposits within blood vessels.

female athlete triad—Three interrelated disorders—disordered eating, menstrual dysfunction, and bone mineral disorders—to which some female athletes are prone.

FEV_{1.0}—See forced expiratory volume in 1 s.

FFA—See free fatty acids.

fiber hyperplasia—An increase in the number of muscle fibers.

fiber hypertrophy—An increase in the size of existing individual muscle fibers.

Fick equation— $\dot{V}O_2 = \dot{Q} \times (a - \bar{v})O_2$ difference.

Fick's law—Law stating that the net diffusion rate of a gas across a fluid membrane is proportional to the difference in partial pressure, proportional to the area of the membrane, and inversely proportional to the thickness of the membrane.

force-velocity relation—The force generated by a muscle is a function of its velocity. Increasing the velocity of contraction while shortening reduces force, while increasing speed of contractions as the muscle is lengthening increases force.

forced expiratory volume in 1 s (FEV_{1.0})—The volume of air exhaled in the first second after maximal inhalation.

Frank-Starling mechanism—The mechanism by which an increased amount of blood in the ventricle causes a stronger ventricular contraction to increase the amount of blood ejected.

free fatty acids (FFA)—The components of fat that are used by the body for metabolism.

free radicals—Univalent (unpaired) oxygen intermediates that leak out of the electron transport chain during metabolic processes and may damage tissues.

free weights—Traditional resistance training modality that uses only barbells, dumbbells, and so on to provide resistance.

frostbite—Tissue damage that occurs during cold exposure because circulation to the skin decreases, in an attempt to retain body heat, to the point that the tissue receives insufficient oxygen and nutrients.

functional sympatholysis—The process in which vasoactive molecules released from active skeletal muscle inhibit sympathetic vasoconstriction in order to increase blood flow to exercising muscle.

gastric emptying—The movement of food mixed with gastric secretions from the stomach into the duodenum.

genomics—The branch of molecular biology concerned with the structure, function, evolution, and mapping of genomes.

genotype—The genetic makeup of an individual.

ghrelin—A hormone secreted from the stomach and pancreas when the stomach is empty in order to stimulate hunger.

glucagon—A hormone released by the pancreas that promotes increased breakdown of liver glycogen to glucose (glycogenolysis) and increased gluconeogenesis.

glucocorticoid—A family of steroid hormones produced by the adrenal cortex that help maintain homeostasis through a variety of effects throughout the body.

gluconeogenesis—The conversion of protein or fat into glucose.

glucose—Six-carbon sugar that is the primary form of carbohydrate used for metabolism.

glycogen—The form of carbohydrate stored in the body, found predominantly in the muscles and liver.

glycogen loading—The manipulation of exercise and diet to optimize the body's glycogen storage.

glycogenolysis—The conversion of glycogen to glucose.

glycogen sparing—Increased reliance on fats for energy production during endurance activity, rather than stores of glycogen.

glycolysis—The breakdown of glucose to pyruvic acid.

Golgi tendon organ—A sensory receptor in a muscle tendon that monitors tension.

graded exercise test (GXT)—An exercise test in which the rate of work is increased gradually in 1 to 3 min increments, usually to the point of fatigue or exhaustion.

graded potential—A localized change (depolarization or hyperpolarization) in the membrane potential.

growth—An increase in the size of the body or any of its parts.

growth hormone—An anabolic agent that stimulates fat metabolism and promotes muscle growth and hypertrophy by facilitating amino acid transport into the cells.

GXT—See graded exercise test.

HACE—See high-altitude cerebral edema.

Haldane transformation—An equation allowing one to calculate the inspired air volume from expired air volume, or expired air volume from inspired air volume.

HAPE—See high-altitude pulmonary edema.

HDL-C—See high-density lipoprotein cholesterol.

health span—The number of years of generally healthy living, free from serious disease or chronic disability.

heart failure—A clinical condition in which the myocardium becomes too weak to maintain adequate cardiac output to meet the body's oxygen demands; heart failure usually results from the heart's being damaged or overworked.

heart murmur—A condition in which abnormal heart valve sounds are detected with the aid of a stethoscope.

heart rate reserve—The difference between maximal heart rate ($HR_{\max}$) and resting heart rate (HR_{rest}).

heat acclimation—See acclimation.

heat cramp—Cramping of the skeletal muscles as a result of excessive dehydration and the associated salt loss.

heat exhaustion—A heat disorder resulting from an inability of the cardiovascular system to meet all the body tissues' needs while also shifting blood to the periphery for cooling, characterized by elevated body temperature, breathlessness, extreme tiredness, dizziness, and rapid pulse.

heatstroke—The most serious heat disorder, resulting from failure of the body's thermoregulatory mechanisms. Heatstroke is characterized by body temperature above 40.5 °C (105 °F), cessation of sweating, and total confusion or unconsciousness; it can lead to death.

hematocrit—The percentage of cells or formed elements in the total blood volume. More than 99% of the cells or formed elements are red blood cells.

hematopoiesis—Increased red blood cell concentration by increased production of cells.

hemoconcentration—A relative (not absolute) increase in the cellular content per unit of blood volume, resulting from a reduction in plasma volume.

hemodilution—An increase in blood plasma, resulting in a dilution of the blood's cellular contents.

hemoglobin—The iron-containing pigment in red blood cells that binds oxygen.

hemorrhagic stroke—Involves bleeding within the brain, which damages nearby brain tissue.

Henry's law—Law stating that gases dissolve in liquids in proportion to their partial pressures, depending also on their solubilities in the specific fluids and on the temperature.

hGH—See human growth hormone.

high-altitude cerebral edema (HACE)—A condition of unknown cause in which fluid accumulates in the cranial cavity at altitude; characterized by mental confusion that can progress to coma and death.

high-altitude pulmonary edema (HAPE)—A condition of unknown cause in which fluid accumulates in the lungs at altitude, interfering with ventilation,

resulting in shortness of breath and fatigue, and characterized by impaired blood oxygenation, mental confusion, and loss of consciousness.

high-density lipoprotein cholesterol (HDL-C)—The cholesterol carried by HDL.

high-intensity interval training (HIIT)—Training that uses short bursts of very intense exercise interspersed with only a few minutes of rest or low-intensity exercise.

high responders—Those individuals within a population that show clear or exaggerated responses or adaptations to a stimulus.

HIIT—See high-intensity interval training.

homeostasis—Maintenance of a constant internal environment.

hormone—A chemical substance produced or released by an endocrine gland and transported by the blood to a specific target tissue.

HR_{max}—See maximum heart rate.

human growth hormone (hGH)—A hormone that promotes anabolism and is believed by some athletes to have ergogenic properties.

hydrostatic pressure—The pressure exerted by a stationary column of fluid in a tube.

hydrostatic weighing—A method of measuring body volume in which a person is weighed while submerged underwater. The difference between the scale weight on land and the underwater weight (corrected for water density) equals body volume. This value must be further corrected to account for any air trapped in the lungs and other parts of the body.

hyperglycemia—An elevated blood glucose level.

hyperinsulinemia—High levels of insulin in the blood.

hyperpolarization—An increase in the electrical potential across a membrane.

hypertension—Abnormally high blood pressure. In adults, hypertension is usually defined as a systolic pressure of 140 mmHg or higher or a diastolic pressure of 90 mmHg or higher.

hypertrophy—Increase in the size or mass of an organ or body tissue. *See also* fiber hypertrophy.

hyperventilation—A breathing rate or tidal volume greater than necessary for normal function.

hypobaric—Referring to an environment, such as that at high altitude, involving low atmospheric pressure.

hypoglycemia—A low blood glucose level.

hyponatremia—A blood sodium concentration below the normal range of 136 to 143 mmol/L.

hypothermia—Low body temperature; any temperature below the given person's normal temperature.

hypoxemia—A decreased oxygen content or concentration within the blood.

hypoxia—A decreased availability of oxygen to the tissues.

IDDM—See insulin-dependent diabetes mellitus.

immune function—The body's normal ability to fight infection and illness with antibodies and lymphocytes.

impaired fasting glucose—A plasma glucose level between 110 and 125 mg/dl following an 8 h fast.

impaired glucose tolerance—An abnormal glucose response to an oral glucose load (glucose tolerance test), sometimes seen as a precursor to diabetes.

independent variable—In an experiment, the variable that is manipulated by the experimenter to determine the response of the dependent variable. Usually plotted on the x-axis.

indirect calorimetry—A method of estimating energy expenditure by measuring respiratory gases.

infancy—The first year of life.

inhibiting factors—Hormones transmitted from the hypothalamus to the anterior pituitary that inhibit release of some other hormones.

inhibitory postsynaptic potential (IPSP)—A hyperpolarization of the postsynaptic membrane caused by an inhibitory impulse.

inspiration—The active process involving the diaphragm and the external intercostal muscles that expands the thoracic dimensions and thus the lungs. The expansion decreases pressure in the lungs, allowing outside air to rush in.

insulation—Resistance to dry heat loss.

insulative acclimation—A pattern of cold acclimation in which enhanced skin vasoconstriction increases peripheral insulation and minimizes heat loss.

insulin—A hormone produced by the β -cells in the pancreas that assists glucose entry into cells.

insulin-dependent diabetes mellitus (IDDM)—One of two major categories of diabetes mellitus that is caused by the inability of the pancreas to produce sufficient insulin as a result of failure of the β -cells in the pancreas. This is also known as type 1 diabetes.

insulin resistance—A physiological condition in which cells fail to respond to the normal actions of the hormone insulin.

insulin sensitivity—An index of the effectiveness of a given insulin concentration on the disposal of glucose.

integrative physiology—The study of organisms as functioning systems of molecules, cells, tissues, and organs with an emphasis on whole-body function.

intercalated disks—Specialized cell junctions in the myocardium where one muscle cell connects to the next.

internal respiration—The exchange of gases between the blood and tissues.

interval-circuit training—Training program that involves rapid movement from one exercise to another around a "circuit" or established set of exercises.

interval training—Repeated, brief, fast-paced exercise bouts with short rest intervals between bouts.

intracellular fluid—The approximately 60% to 65% of total body water that is contained in the cells.

IPSP—See inhibitory postsynaptic potential.

irisin—A muscle-derived hormone that signals conversion of white adipose tissue to brown adipose tissue.

ischemia—A temporary deficiency of blood to a specific area of the body.

ischemic stroke—Brain tissue damage resulting from insufficient oxygen supply to an area of the brain. May be caused by narrowing or blockage of blood vessels supplying the area.

isokinetic training—Resistance training in which the rate of movement is kept constant through the range of motion.

isometric training—Resistance training involving a static action.

kilocalorie (kcal)—The equivalent of 1,000 calories. *See also* calorie.

Krebs cycle—A series of chemical reactions that involve the complete oxidation of acetyl CoA and produce 2 mol of ATP (energy) along with hydrogen and carbon, which combine with oxygen to form H₂O and CO₂.

lactate threshold—The point during exercise of increasing intensity at which blood lactate begins to accumulate above resting levels, where lactate clearance is no longer able to keep up with lactate production.

LDL-C—See low-density lipoprotein cholesterol.

length-tension relation—The tension developed by a muscle is a function of its length. The highest force can be generated when there is optimal overlap between actin and myosin filaments.

leptin—A hormone primarily secreted by fat cells that acts on receptors in the hypothalamus to decrease hunger.

lipogenesis—The process of converting protein into fatty acids.

lipolysis—The process of breaking down triglyceride to its basic units to be used for energy.

lipoprotein lipase—The enzyme that breaks down triglycerides to free fatty acids and glycerol, allowing the free fatty acids to enter the cells for use as a fuel or for storage.

lipoproteins—The proteins that carry the blood lipids.

longevity—The length of a person's life.

longitudinal research design—A research design in which subjects are tested initially and then one or more times later to directly measure changes over time resulting from a given intervention.

long, slow distance (LSD) training—Endurance training involving long, slow distances.

low-density lipoprotein cholesterol (LDL-C)—The cholesterol carried by LDL.

lower body (gynoid) obesity—Obesity that follows the typically female pattern of fat storage, in which fat is stored primarily in the lower body, particularly in the hips, buttocks, and thighs.

low-intensity aerobic exercise—Aerobic exercise performed at low intensity, theoretically to cause the body to burn a higher percentage of fat.

low responders—Those individuals within a population who show little or no response or adaptation to a stimulus.

macrominerals—Those minerals of which the body needs more than 100 mg per day.

MAP—See mean arterial pressure.

maturation—The process by which the body takes on the adult form and becomes fully functional. It is often defined by the system or function being considered.

maximal expiratory ventilation ($\dot{V}_{E\max}$)—The highest ventilation that can be achieved during exhaustive exercise.

maximal oxygen uptake ($\dot{V}O_{2\max}$)—The maximal capacity for oxygen consumption by the body during maximal exertion. It is also known as aerobic power, maximal oxygen intake, maximal oxygen consumption, and cardiorespiratory endurance capacity.

maximal voluntary ventilation—The maximal capacity to move air into and out of the lungs, usually measured for 12 s and extrapolated to a per minute value.

maximum heart rate ($HR_{\max}$)—The highest heart rate value attainable during an all-out effort to the point of exhaustion.

mean arterial pressure (MAP)—The average pressure exerted by the blood as it travels through the arteries. It is estimated as follows: $MAP = DBP + [0.333 \times (SBP - DBP)]$.

mechanoreceptors—An end organ that responds to changes in mechanical stress, such as stretch, compression, or distension.

menarche—The onset of menstruation; the first menses.

menopause—The permanent cessation of menses in women, typically occurring between ages 45 and 55; defined as occurring 12 months after the final menstrual period.

menses—The menstrual or flow phase of the menstrual cycle.

menstrual cycle—The cycle of uterine changes, averaging 28 days and consisting of the menstrual (flow) phase, the proliferative phase, and the secretory phase.

menstrual dysfunction—Disruption of the normal menstrual cycle; includes oligomenorrhea, primary amenorrhea, and secondary amenorrhea.

metabolic acclimation—A pattern of cold acclimation involving increased metabolic heat production through enhanced nonshivering and shivering thermogenesis.

metabolic equivalent (MET)—A unit used to estimate the metabolic cost (oxygen consumption) of physical activity. One MET equals the resting metabolic rate of approximately $3.5 \text{ ml of } O_2 \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$.

metabolic syndrome—A term that has been used to link coronary artery disease, hypertension, type 2 diabetes, and upper body obesity to insulin resistance and hyperinsulinemia. This syndrome has also been referred to as syndrome X and the civilization syndrome.

metabolism—All energy-producing and energy-using processes within the body.

microminerals (trace elements)—The minerals of which the body needs less than 100 mg per day.

mineralocorticoids—Steroid hormones released from the adrenal cortex that are responsible for electrolyte balance within the body—for example, aldosterone.

mitochondria—Cellular organelles that generate ATP through oxidative phosphorylation.

mitochondrial oxidative enzymes—Oxidative enzymes located in the mitochondria.

mode—Type of exercise.

motor nerves (motor neurons)—Efferent nerves that carry impulses to skeletal muscle.

motor neurons—See motor nerves.

motor reflex—An involuntary motor response to a given stimulus.

motor unit—The motor nerve and the group of muscle fibers it innervates.

mTOR—Enzyme that controls the rate of protein synthesis within the myofibrils after resistance training.

muscle fiber—An individual muscle cell.

muscle pump—The rhythmic mechanical compression of the veins that occurs during skeletal muscle contraction in many types of movement and exercise—for example, during walking and running—and assists the return of blood to the heart.

muscle spindle—A sensory receptor located in the muscle that senses how much the muscle is stretched.

muscular endurance—The ability of a muscle to resist fatigue.

musculoskeletal system—Body system composed of the skeleton and skeletal muscles that allows, supports, and helps control human movement.

myelination—The process of acquiring a myelin sheath.

myelin sheath—The outer covering of a myelinated nerve fiber, formed by a fatlike substance called myelin.

myocardial infarction—Death of heart tissue that results from insufficient blood supply to part of the myocardium.

myocardium—The muscle of the heart.

myofibril—The contractile element of skeletal muscle.

myoglobin—A compound similar to hemoglobin, but found in muscle tissue, that carries oxygen from the cell membrane to the mitochondria.

myosin—One of the proteins that form filaments that produce muscle action.

myosin cross-bridge—The protruding part of a myosin filament. It includes the myosin head, which binds to an active site on an actin filament to produce a power stroke that causes the filaments to slide across each other.

nebulin—A giant protein that coextends with actin and appears to play a regulatory role in mediating actin and myosin interactions.

negative feedback—A decrease in the amount or effect of a molecule or substance by its influence on the process giving rise to it, such as when a high level of a particular hormone in the blood may inhibit further secretion of that hormone.

nerve impulse—The electrical signal conducted along a neuron, which can be transmitted to another neuron or an end organ such as a group of muscle fibers.

neuromuscular junction—The site at which a motor neuron communicates with a muscle fiber.

neuron—A specialized cell in the nervous system responsible for generating and transmitting nerve impulses.

neurotransmitter—A chemical used for communication between a neuron and another cell.

nitric oxide (NO)—Important cellular signaling molecule involved in many physiological processes, including dilation of arterioles.

nonessential amino acids—The 11 or 12 amino acids that the body synthesizes.

nonshivering thermogenesis—The stimulation of metabolism by the sympathetic nervous system to generate more metabolic heat.

nonsteroid hormones—Hormones derived from protein, peptides, or amino acids that cannot easily cross cell membranes.

norepinephrine—A catecholamine released from the adrenal medulla that, along with epinephrine, prepares the body for a fight-or-flight response. It is also a neurotransmitter. *See also* catecholamines.

obesity—An excessive amount of body fat, generally defined as more than 25% in men and 35% in women; a body mass index of 30 or greater.

oligomenorrhea—Abnormally infrequent or scant menstruation.

oncotic pressure—The pressure exerted by the concentration of proteins in a solution, drawing water from regions with lower oncotic pressures.

osmolality—The number of solutes (such as electrolytes) dissolved in a fluid divided by the weight of that fluid; usually expressed in units of osmoles (or milliosmoles) per kilogram.

osmolarity—The number of solutes (such as electrolytes) dissolved in a fluid divided by the volume of that fluid; usually expressed in units of osmoles (or milliosmoles) per liter.

ossification—The process of bone formation.

osteopenia—The loss of bone mass with aging.

osteoporosis—Decreased bone mineral content that increases bone porosity.

overreaching—A systematic attempt to intentionally overstress the body, allowing the body to adapt to the training stimulus above and beyond the adaptation attained during a period of acute overload.

overtraining—The attempt to do more work than can be physically tolerated.

overtraining syndrome—A condition brought on by overtraining and characterized by performance decrements and a general breakdown in physiological function.

overweight—Body weight that exceeds the normal or standard weight for a particular individual based on sex, height, and frame size; a BMI of 25.0 to 29.9.

oxidative phosphorylation—Mitochondrial process that uses oxygen and high-energy electrons to produce ATP and water.

oxidative system—The body's most complex energy system, which generates energy by disassembling fuels with the aid of oxygen and has a very high energy yield.

oxygen cascade—The progressively dropping partial pressures of oxygen from dry ambient air as oxygen flows to the tissues and into the venous circulation draining those tissues.

oxygen deficit—The difference between the oxygen required for a given exercise intensity (steady state) and the actual oxygen consumption.

oxygen diffusion capacity—The rate at which oxygen diffuses from one place to another.

oxygen transport system—The components of the cardiovascular and respiratory systems involved in transporting oxygen.

partial pressure—The pressure exerted by an individual gas in a mixture of gases.

partial pressure of oxygen (PO_2)—The pressure exerted by oxygen in a mixture of gases.

pathophysiology—The physiology of a specific disease or disorder.

P_b —See barometric pressure.

PCr—See phosphocreatine.

PDGF—See platelet-derived growth factor.

peak oxygen uptake ($\dot{V}O_{2peak}$)—The highest oxygen uptake achieved during a graded exercise test when a subject reaches volitional fatigue before a plateau occurs in the $\dot{V}O_2$ response (the criterion for a true $\dot{V}O_{2max}$).

pericardium—A double-layered outer covering of the heart.

perimysium—The connective tissue sheath surrounding each muscle fasciculus.

periodization—The breaking down of an entire season's training program for sport into smaller periods or training units.

peripheral blood flow—Blood flow to the extremities and the skin.

peripheral nervous system (PNS)—That section of the nervous system through which motor nerve impulses are transmitted from the brain and spinal cord to the

periphery and sensory nerve impulses are transmitted from the periphery to the brain and spinal cord.

peripheral vascular disease—Diseases of the systemic arteries and veins, especially those to the extremities, that impede adequate blood flow.

peripheral vasoconstriction—See vasoconstriction.

PFK—See phosphofructokinase.

phenotype—Observable characteristics of an individual resulting from the interaction of his or her genotype with the environment.

phosphocreatine (PCr)—An energy-rich compound that plays a critical role in providing energy for muscle action by maintaining ATP concentration.

phosphofructokinase (PFK)—A key rate-limiting enzyme of the anaerobic glycolytic energy system.

phosphorylation—The addition of a phosphate group (PO_4) to a molecule.

physical maturity—The point at which the body has attained the adult physical form.

physiology—The study of the function of organisms.

placebo effect—An effect produced by the subject's expectations after administration of an inactive substance (placebo).

placebo group—The group in an intervention study given a placebo rather than a test substance.

platelet-derived growth factor (PDGF)—A family of molecules released from platelets in the blood that help to heal wounds, repair damage to blood vessel walls, and help blood vessels grow.

plaque—A buildup of lipids, smooth muscle cells, connective tissue, and debris that forms at the site of injury to an artery.

plasmalemma—Plasma membrane, the selectively permeable lipid bilayer coated by proteins that composes the outer layer of a cell.

plyometrics—A type of dynamic-action resistance training based on the theory that use of the stretch reflex during jumping will recruit additional motor units.

PNS—See peripheral nervous system.

PO_2 —See partial pressure of oxygen.

POAH—See preoptic-anterior hypothalamus.

polycythemia—Increased red blood cells.

power—The rate of performing work; the product of force and velocity. The rate of transformation of metabolic potential energy to work or heat.

power stroke—The tilting of the myosin head, caused by a strong intermolecular attraction between the myosin cross-bridge and the myosin head, that causes the actin and myosin filaments to slide across each other.

prediabetes—A term used to define those who have impaired fasting glucose, impaired glucose tolerance, or both, but are not truly diabetic.

predictive value of an abnormal exercise test—The accuracy with which abnormal test results reflect the presence of a disease.

pregnancy—The state of carrying an embryo or fetus in the body.

preload—The degree to which the myocardium is stretched before it contracts, determined by factors such as central blood volume.

premature ventricular contraction (PVC)—A common cardiac arrhythmia that results in the feeling of skipped or extra beats caused by impulses originating outside the SA node.

preoptic-anterior hypothalamus (POAH)—The area of the midbrain that is the primary controller of thermoregulatory function.

primary aging—The effect of chronological age alone, i.e., in the absence of other comorbidities that often accompany aging.

primary amenorrhea—The absence of menarche (the beginning of menstruation) beyond age 18.

primary risk factors—Risk factors that have been conclusively shown to have a strong association with a certain disease. Primary risk factors for coronary artery disease include smoking, hypertension, high blood lipid levels, obesity, and physical inactivity.

principle of individuality—The theory that any training program must consider the specific needs and abilities of the individual for whom it is designed.

principle of orderly recruitment—The theory that motor units generally are activated on the basis of a fixed order of recruitment, in which the motor units within a given muscle appear to be ranked according to the size of the motor neuron.

principle of periodization—The gradual cycling of specificity, intensity, and volume of training to achieve peak levels of fitness for competition. Also called the principle of variation.

principle of progressive overload—The theory that, to maximize the benefits of a training program, the training stimulus must be progressively increased as the body adapts to the current stimulus.

principle of reversibility—The theory that a training program must include a maintenance plan to ensure that the gains from training are not lost.

principle of specificity—The theory that a training program must stress the physiological systems critical for optimal performance in a given sport to achieve desired training adaptations in that sport.

principle of variation—The systematic process of changing one or more variables in an exercise training program—mode, volume, or intensity—over time to allow for the training stimulus to remain challenging and effective. Also called the principle of periodization.

prostaglandins—Substances derived from a fatty acid that act as hormones at the local level.

protein—A class of nitrogen-containing compounds formed by amino acids.

pseudoephedrine—A sympathomimetic amine that is used in over-the-counter medications primarily as a decongestant and in the illicit manufacturing of methamphetamine.

puberty—The point at which a person becomes physiologically capable of reproduction.

pulmonary diffusion—The exchange of gases between the lungs and the blood.

pulmonary ventilation—The movement of gases into and out of the lungs.

Purkinje fibers—The terminal branches of the AV bundle that transmit impulses through the ventricles six times faster than through the rest of the cardiac conduction system.

PVC—See premature ventricular contraction.

$\dot{Q}$ —See cardiac output.

radiation (R)—The transfer of heat through electromagnetic waves.

rate coding—Refers to the frequency of impulses sent to a muscle. Increased force can be generated through increase in either the number of muscle fibers recruited or the rate at which the impulses are sent. Also called frequency coding.

rate-limiting enzyme—An enzyme found early in a metabolic pathway that determines the rate of the pathway.

rate-pressure product (RPP)—The mathematical product of heart rate $\times$ systolic blood pressure. Also called the double product.

rating of perceived exertion (RPE)—A person's subjective assessment of how hard he or she is exercising.

rehabilitation programs—Programs designed to reestablish health or fitness following a disability or illness.

relative body fat—The ratio of fat mass to total body mass, expressed as a percentage.

releasing factors—Hormones transmitted from the hypothalamus to the anterior pituitary that promote release of some other hormones.

renin—An enzyme formed by the kidneys to convert a plasma protein called angiotensinogen into angiotensin II. See *also* renin-angiotensin-aldosterone mechanism.

renin-angiotensin-aldosterone mechanism—The mechanism involved in renal control of blood pressure. The kidneys respond to decreased blood pressure or blood flow by forming renin, which converts angiotensinogen into angiotensin I, which is finally converted to angiotensin II. Angiotensin II constricts arterioles and triggers aldosterone release.

RER—See respiratory exchange ratio.

residual volume (RV)—The amount of air that cannot be exhaled from the lungs.

resistance training—Training designed to increase strength, power, and muscular endurance.

respiratory alkalosis—A condition in which increased carbon dioxide clearance allows blood pH to increase.

respiratory centers—Autonomic centers located in the medulla oblongata and the pons that establish breathing rate and depth.

respiratory exchange ratio (RER)—The ratio of carbon dioxide expired to oxygen consumed at the level of the lungs.

respiratory membrane—The membrane separating alveolar air and blood, composed of the alveolar wall, the capillary wall, and their basement membranes.

respiratory pump—Passive movement of blood through the central circulation as a function of pressure changes during breathing.

resting heart rate (RHR)—The heart rate at rest, averaging 60 to 80 beats/min.

resting membrane potential (RMP)—The potential difference between the electrical charges inside a cell and outside the cell, caused by a separation of charges across the membrane.

resting metabolic rate (RMR)—The body's metabolic rate early in the morning following an overnight fast and 8 h of sleep. Determining RMR does not require sleeping overnight in a laboratory or clinical facility. *See also* basal metabolic rate.

rheumatic heart disease—A form of valvular heart disease involving a streptococcal infection that has caused acute rheumatic fever, typically in children between ages 5 and 15.

RHR—See resting heart rate.

risk factor—A predisposing factor statistically linked to the development of a disease, such as coronary artery disease.

RMP—See resting membrane potential.

RMR—See resting metabolic rate.

RPE—See rating of perceived exertion.

RV—See residual volume.

saltatory conduction—The means of rapid nerve impulse conduction along myelinated neurons.

SA node—See sinoatrial node.

sarcolemma—A muscle fiber's cell membrane.

sarcomere—The basic functional unit of a myofibril.

sarcopenia—The loss of muscle mass associated with aging.

sarcoplasm—The gelatin-like cytoplasm in a muscle fiber.

sarcoplasmic reticulum (SR)—A longitudinal system of tubules that is associated with the myofibrils and stores calcium for muscle action.

satellite cells—Immature cells that can develop into mature cell types, such as myoblasts.

SBP—See systolic blood pressure.

secondary amenorrhea—The cessation of menstruation in a woman with previously normal menstrual function.

second messenger—A substance inside a cell that acts as a messenger after a nonsteroid hormone binds to receptors outside the cell.

sensitivity—A test's ability to correctly identify subjects who fit the criteria being tested, such as coronary artery disease.

sensory (afferent) nerves—Afferent nerves that carry impulses toward the central nervous system from the periphery.

sensory-motor integration—The process by which the sensory and motor systems communicate and coordinate with each other.

sex-specific differences—True physiological differences between females and males.

shivering—A rapid, involuntary cycle of contraction and relaxation of skeletal muscles that generates heat.

sickle cell trait—Inheritance of one gene for the disease sickle cell anemia leads to a condition in which those afflicted are at an increased risk for several pathologies that are exacerbated by exercise and dehydration.

single-fiber contractile velocity (V_o)—The rate at which an individual muscle cell can shorten and develop tension.

sinoatrial (SA) node—A group of specialized myocardial cells, located in the wall of the right atrium, that control the heart's rate of contraction; the pacemaker of the heart.

size principle—Principle asserting that the size of the motor neuron dictates the order of motor unit recruitment, with small-sized motor neurons being recruited first.

skinfold fat thickness—The most widely applied field technique used to estimate body density, relative body fat, and fat-free mass. It involves measurement with calipers of the skinfold fat at one or more sites.

sliding filament theory—A theory explaining muscle action: A myosin cross-bridge attaches to an actin filament, and then the power stroke drags the two filaments past one another.

sodium-potassium pump—An enzyme called $\text{Na}^+\text{-K}^+\text{-ATPase}$, which maintains the resting membrane potential in disequilibrium at -70 mV.

specificity—A test's ability to correctly identify subjects who do not fit the criteria being tested.

specificity of training—The principle that physiological adaptations in response to physical training are highly specific to the nature of the training activity. To maximize benefits, training should be carefully matched to an athlete's specific performance needs.

spirometry—The measurement of lung volumes and capacities.

sport physiology—The application of the concepts of exercise physiology to training athletes and enhancing sport performance.

spot reduction—The practice of exercising a specific area of the body, theoretically to reduce locally stored fat.

SR—See sarcoplasmic reticulum.

static-contraction resistance training—Resistance training that emphasizes static muscle action. Also known as isometric resistance training.

static (isometric) muscle contraction—Action in which the muscle contracts without moving, generating force while its length remains static (unchanged). Also known as isometric action.

steady-state heart rate—A heart rate that is maintained constant at submaximal levels of exercise when the rate of work is held constant.

steroid hormones—Hormones with chemical structures similar to cholesterol that are lipid soluble and that diffuse through cell membranes.

strength—The ability of a muscle to exert force—generally the maximal ability.

stroke—A cerebral vascular accident, a condition in which blood supply to some part of the brain is impaired, typically caused by infarction or hemorrhage, so that the tissue is damaged.

stroke volume (SV)—The amount of blood ejected from the left ventricle during contraction; the difference between the end-diastolic volume and the end-systolic volume.

submaximal endurance—The average absolute power output a person can maintain during a fixed period of time on a cycle ergometer, or the average speed or velocity a person can maintain during a fixed period of time. Generally, these tests last at least 30 min but usually not more than 90 min.

substrate—Basic fuel source, such as carbohydrate, protein, and fat.

summation—The summing of all individual changes in a neuron's membrane potential.

SV—See stroke volume.

synapse—The junction between two neurons.

systolic blood pressure (SBP)—The greatest arterial blood pressure, resulting from systole (the contracting phase of the heart).

T₃—See triiodothyronine.

T₄—See thyroxine.

tachycardia—A resting heart rate greater than 100 beats/min.

tapering—A reduction in training intensity before a major competition to give the body and mind a break from the rigors of intense training.

taper period—A period during which training intensity is reduced, allowing time for tissue damage from intense training to heal and for the body's energy reserves to be fully replenished.

target cells—Cells that possess specific hormone receptors.

TEA—See thermic effect of activity.

TEM—See thermic effect of a meal.

testosterone—The predominant male sex hormone.

tetanus—Highest tension developed by a muscle in response to stimulation of increasing frequency.

therapeutic use exemption—An exemption granted by the governing body of a sport that allows an athlete to use an otherwise banned substance if it is needed to treat a medical condition.

thermal stress—Stress imposed on the body by external temperature.

thermic effect of activity (TEA)—The energy expended in excess of the resting metabolic rate to accomplish a given task or activity.

thermic effect of a meal (TEM)—The energy expended in excess of resting metabolic rate associated with digestion, absorption, transport, metabolism, and storage of ingested food.

thermoreceptors—Sensory receptors that detect changes in body temperature and external temperature and relay this information to the hypothalamus. Also called thermoceptors.

thermoregulation—The process by which the thermoregulatory center, located in the hypothalamus, readjusts body temperature in response to small deviations from the set point.

thirst mechanism—A neural mechanism that triggers thirst in response to dehydration.

THR—See training heart rate.

threshold—A minimum amount of stimulus needed to elicit a response. Also, the minimum depolarization required to produce an action potential in neurons.

thyrotropin (TSH)—A hormone secreted by the anterior lobe of the pituitary gland that promotes the release of thyroid hormones.

thyroxine (T₄)—A hormone secreted by the thyroid gland that increases the rate of cellular metabolism and the rate and contractility of the heart.

tidal volume—The amount of air inspired or expired during a normal breathing cycle.

titin—A protein that positions the myosin filament to maintain equal spacing between actin filaments.

TLC—See total lung capacity.

total lung capacity (TLC)—The sum of vital capacity and residual volume.

total peripheral resistance (TPR)—The resistance to the flow of blood through the entire systemic circulation.

trace elements—See microminerals.

training effect—Physiological adaptation to repeated bouts of exercise.

training heart rate (THR)—A heart rate goal established by using the heart rate equivalent of a desired percentage of $\dot{V}O_{2\max}$. For example, if a training level of

75% $\dot{V}O_{2\max}$ is desired, 75% of $\dot{V}O_{2\max}$ is calculated, and the heart rate corresponding to this $\dot{V}O_2$ is selected as the THR.

transient hypertrophy—The “pumping up” of muscle that happens during a single exercise bout, resulting mainly from fluid accumulation in the interstitial and intracellular spaces of the muscle.

translational physiology—The processes by which basic research findings are extended to the clinical research setting, then to the realm of clinical practice, and finally to health policy.

transverse tubules (T-tubules)—Extensions of the sarcolemma (plasma membrane) that pass laterally through the muscle fiber, allowing nutrients to be transported and nerve impulses to be transmitted rapidly to individual myofibrils.

treadmill—An ergometer in which a motor and pulley system drive a large belt that a person can either walk or run on.

triglycerides—The body’s most concentrated energy source and the form in which most fats are stored in the body.

triiodothyronine (T₃)—A hormone released by the thyroid gland that increases the rate of cellular metabolism and the rate and contractility of the heart.

tropomyosin—A tube-shaped protein that twists around actin strands, fitting into the groove between them.

troponin—A complex protein attached at regular intervals to actin strands and tropomyosin.

TSH—See thyrotropin.

twitch—The smallest contractile response of a muscle fiber or a motor unit to a single electrical stimulus.

type 1 diabetes—A type of diabetes mellitus that generally has a sudden onset during childhood or young adulthood and leads to almost total insulin deficiency, usually requiring daily insulin injections. Also known as insulin-dependent diabetes mellitus (IDDM) or juvenile-onset diabetes.

type 2 diabetes—A type of diabetes mellitus in which disease onset is more gradual and the causes are more difficult to establish than in type 1 diabetes. Type 2 diabetes is characterized by impaired insulin secretion, impaired insulin action, or excessive glucose output from the liver. Also known as non-insulin-dependent diabetes mellitus (NIDDM).

type I fiber—A type of muscle fiber that has a high oxidative and a low glycolytic capacity, associated with endurance-type activities.

type II fiber—A type of muscle fiber with a low oxidative capacity and a high glycolytic capacity; associated with speed or power activities.

undertraining—The type of training an athlete would undertake between competitive seasons or during active rest. Generally, physiological adaptations will be minor, and there will be no improvement in performance.

upper body (android) obesity—Obesity that follows the typically male pattern of fat storage, in which fat is stored primarily in the upper body, particularly in the abdomen.

upregulation—An increased cellular sensitivity to a hormone, often caused by increased hormone receptors.

Valsalva maneuver—The process of holding the breath and attempting to compress the contents of the abdominal and thoracic cavities, causing increased intra-abdominal and intrathoracic pressure.

valvular heart disease—A disease involving one or more of the heart valves. Rheumatic heart disease is one example.

variable-resistance training—A technique that allows variation in the resistance applied throughout the range of motion in an attempt to match the ability of the muscle or muscle groups to apply force at any specific point in the range of motion.

vasoconstriction—The constriction or narrowing of blood vessels.

vasodilation—The dilation or widening of blood vessels.

vasopressin—See antidiuretic hormone.

VC—See vital capacity.

$\dot{V}_{E\max}$ —See maximal expiratory ventilation.

$\dot{V}_E/\dot{V}\text{CO}_2$ —See ventilatory equivalent for carbon dioxide.

$\dot{V}_E/\dot{V}\text{O}_2$ —See ventilatory equivalent for oxygen.

veins—Blood vessels that transport blood back to the heart.

ventilatory equivalent for carbon dioxide ($\dot{V}_E/\dot{V}\text{CO}_2$)—The ratio of the volume of air ventilated ($\dot{V}_E$) to the amount of carbon dioxide produced ($\dot{V}\text{CO}_2$).

ventilatory equivalent for oxygen ($\dot{V}_E/\dot{V}\text{O}_2$)—The ratio between the volume of air ventilated ($\dot{V}_E$) and the amount of oxygen consumed ($\dot{V}\text{O}_2$); indicates breathing economy.

ventilatory threshold—Older name for the ventilatory breakpoint.

ventricular fibrillation—A serious cardiac arrhythmia in which the contraction of the ventricular tissue is uncoordinated, affecting the heart's ability to pump blood. See *also* ventricular tachycardia.

ventricular tachycardia—A serious cardiac arrhythmia consisting of three or more consecutive premature ventricular contractions. See *also* premature ventricular contraction and ventricular fibrillation.

venules—Small vessels that transport blood from the capillaries to the veins and then back to the heart.

vital capacity (VC)—The maximal volume of air expelled from the lungs after maximal inhalation.

vitamin—One of a group of unrelated organic compounds that perform specific functions to promote growth and to maintain health. Vitamins act primarily as catalysts in chemical reactions.

$\dot{V}O_2$ —The volume of oxygen consumed per minute.

$\dot{V}O_2$ **drift**—A slow increase in $\dot{V}O_2$ during prolonged submaximal exercise at a constant power output.

$\dot{V}O_{2max}$ —See maximal oxygen uptake.

wet-bulb globe temperature (WBGT)—A measurement of temperature that simultaneously accounts for conduction, convection, evaporation, and radiation, providing a single temperature reading to estimate the cooling capacity of the surrounding environment. The apparatus for measuring WBGT consists of a dry bulb, a wet bulb, and a black globe.

windchill—A chill factor created by the increase in the rate of heat loss via convection and conduction caused by wind.